灵熙智能 · AI进阶课

第一讲 · 60分钟

AI 简史

从规则、学习到会行动的 Agent



这不是模型名单，而是一部关于期待、突破、失望与重新出发的历史。

历史主线　3个经典思想　从过去看未来

全景 · 一条潮起潮落的历史

AI不是一路胜利，而是经历了三次“想象扩大—现实碰壁—条件成熟”

1940s—50s AI成为问题 | 1956—70s 符号主义兴起 | 1970s—90s 寒冬与专家系统 | 1997—2012 机器学习复兴 | 2012→ 深度学习拐点

1940s—50s
提出问题
机器能否表现出智能?

→

1956—70s
第一次热潮
搜索与符号规则

→

1970s—90s
寒冬与复兴
承诺受挫，专家系统再起

→

1997—2012
学习路线回归
数据、算力、神经网络

→

2012→
深度学习时代
规模化学习成为主流

先记住历史规律：演示领先于交付，期待就会变成泡沫；算法、数据、算力和产品同时成熟，突破才会留下。

AI诞生前，三种思想先会合：计算、神经元与可观察行为

1943

神经元可以被抽象成计算单元

McCulloch与Pitts用逻辑模型描述神经元，埋下神经网络路线。

1946

电子计算机把“自动计算”变成现实

ENIAC等机器让复杂计算从纸面想象变成工程能力。

1950

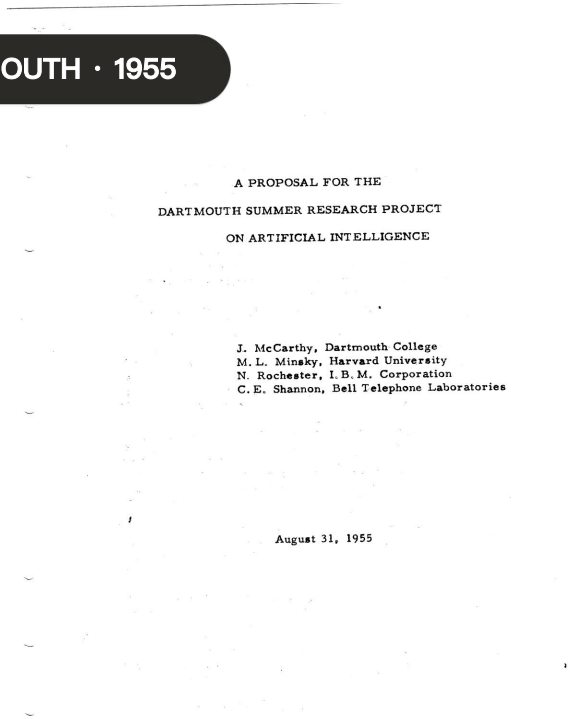
图灵把“会思考吗”改写成行为实验

不争论灵魂，先观察机器能否在对话中表现得难以区分。

领域还没有名字，但关键前提已经齐了：智能可以被描述、计算、观察。

达特茅斯会议给AI命名，也许下了一个影响七十年的承诺

DARTMOUTH · 1955



A PROPOSAL FOR THE

DARTMOUTH SUMMER RESEARCH PROJECT

ON ARTIFICIAL INTELLIGENCE

J. McCarthy, Dartmouth College
M. L. Minsky, Harvard University
N. Rochester, I. B. M. Corporation
C. E. Shannon, Bell Telephone Laboratories

August 31, 1955

核心假设

学习和智能的每个方面，原则上都能被精确描述，让机器模拟。

这句话带来两条路线

写出来
逻辑、规则、搜索

学出来
神经网络、自我改进

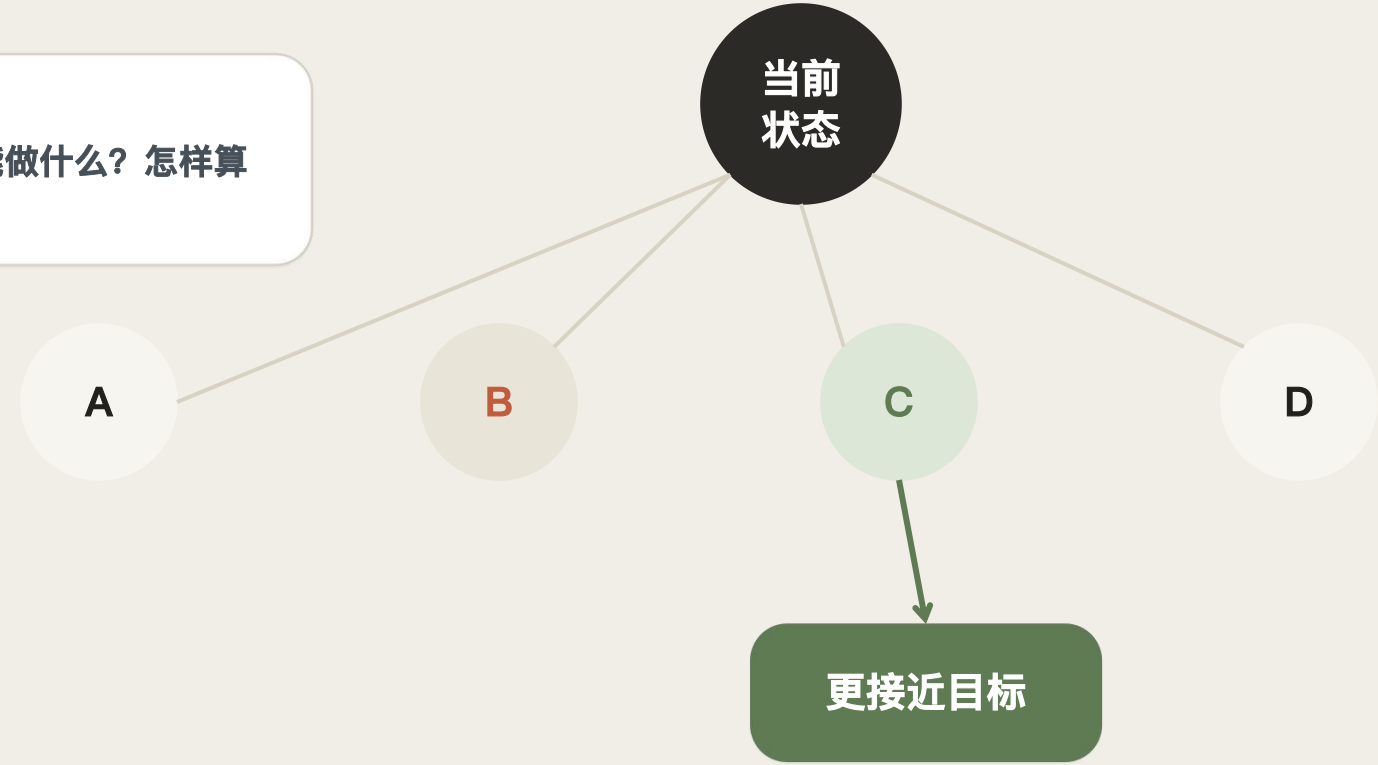
第一次热潮先选择了更容易在当时计算机上实现的路线：把智能写成符号和规则。

承上 | 智能已经可以被描述、计算与观察

下一幕 | 第一次热潮为什么最先押注搜索与符号规则04

早期AI的直觉很朴素：如果答案藏在可能路径里，就让机器去找

把任务写成：
现在在哪里？能做什么？怎样算完成？



只需记住

搜索不是理解世界，而是在被人定义好的世界里找路。

这个思想今天仍在：导航、博弈、规划、推理与Agent，都需要在候选行动中选择下一步。

第一次热潮 · 1956—1969

机器会证明定理、解代数题、玩游戏——通用智能似乎只差几年

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点

1956

Logic Theorist

能证明《数学原理》中的定理，成为早期符号AI标志。

1959

General Problem Solver

研究者尝试用统一搜索方法解决不同形式的问题。

1960s

机器翻译与机器人

政府、大学和企业相信语言与现实行动很快会被攻克。

真正推动泡沫的不是某个算法，而是把窄任务的成功外推成“通用智能即将到来”。

1966 · ELIZA

一个简单规则程序，让人第一次真切感到“机器好像理解我”

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点

对话

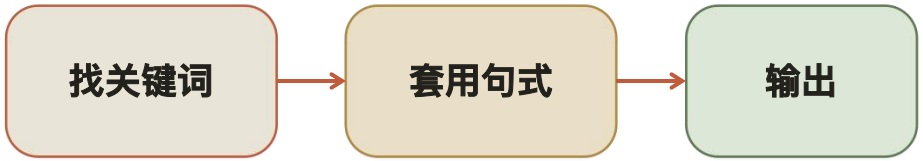
人：我最近总是很焦虑。

机器：你为什么会焦虑?

人：别人不理解我。

机器：你为什么认为别人不理解你?

内部机制



历史意义

ELIZA没有理解用户，却揭示了一个长期现象：人类会主动把连贯语言解释成理解。

这也是今天判断AI仍要警惕的错觉：像理解，不等于稳定理解。

第一次寒冬 · 1966—1980

现实比实验室复杂得多：组合爆炸、常识缺失、算力不足同时暴露

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点

搜索空间爆炸

分支每增加一层，候选数量成倍增长。

开放世界没有边界

真实语言、视觉和常识无法被完整枚举。

算力与数据太少

方法即使可行，也无法支撑真实规模。

承诺过早

机器翻译等项目未达到资助方预期。

寒冬不是“AI消失”，而是社会第一次发现：会做演示，距离稳定解决现实问题还很远。

神经网络曾在1958年引发轰动，又因能力和条件不足陷入低谷

PSYCHOLOGICAL REVIEW · 1958

INFORMATION STORAGE AND ORGANIZATION IN THE BRAIN[1]

F. ROSENBLATT

Cornell Aeronautical Laboratory

If we are eventually to understand the capability of higher organisms for perceptual recognition, generalization, recall, and thinking, we must first have answers to three fundamental questions:

1. How is information about the physical world sensed, or detected, by the biological system?
2. In what form is information stored, or remembered?
3. How does information contained in storage, or in memory, influence recognition and behavior?

The first of these questions is in the province of sensory physiology, and is the only one for which appreciable understanding has been achieved. This article will be concerned primarily with the second and third questions, which are still subject to a vast amount of speculation, and where the few relevant facts currently supplied by neurophysiology have not yet been integrated into an acceptable theory.

With regard to the second question, two alternative positions have been maintained. The first suggests that storage of sensory information is in the form of coded representations or images, with some sort of one-to-one mapping between the sensory stimulus and the stored pattern. According to this hypothesis, if one understood the code or "wiring diagram" of the nervous system, one should, in principle, be able to discover exactly what an organism remembers by reconstructing the original sensory patterns from the "memory traces" which they have left, much as we might develop a photographic negative, or translate the pattern of electrical charges in the "memory" of a digital computer. This hypothesis is appealing in its simplicity and ready intelligibility, and a large family of theoretical brain models has been developed around the idea of a coded, representational memory (**2**, **3**, **9**, **14**). The alternative approach, which stems from the tradition of British empiricism, hazards the guess that the images of stimuli may never really be recorded at all, and that the central nervous system simply acts as an intricate switching network, where retention takes the form of new connections, or pathways, between centers of activity. In many of the more recent developments of this position (Hebb's "cell assembly," and Hull's "cortical anticipatory goal response," for example) the "responses" which are associated to stimuli may be entirely contained within the CNS itself. In this case the response represents an "idea" rather than an action. The important feature of this approach is that there is never any simple mapping of the stimulus into memory, according to some code which would permit its later reconstruction. Whatever in-

[1] The development of this theory has been carried out at the Cornell Aeronautical Laboratory, Inc., under the sponsorship of the Office of Naval Research, Contract Nonr-2381(00). This article is primarily an adaptation of material reported in Ref. 15, which constitutes the first full report on the program.

386

为什么令人兴奋

机器不必记住每条规则，
可以从样本中调整连接权重。

为什么又沉寂

单层模型能力有限；多层网络缺少成熟训练方法；
数据、芯片和工程条件都没有准备好。

历史教训：一个正确方向，也可能因为时代条件不成熟而等待几十年。

灵熙智能 LINGXI INTELLIGENT
灵熙智能内部培训材料

第二次热潮 · 1970s—1980s

AI不再承诺“什么都懂”，而是用专家系统解决一个窄领域

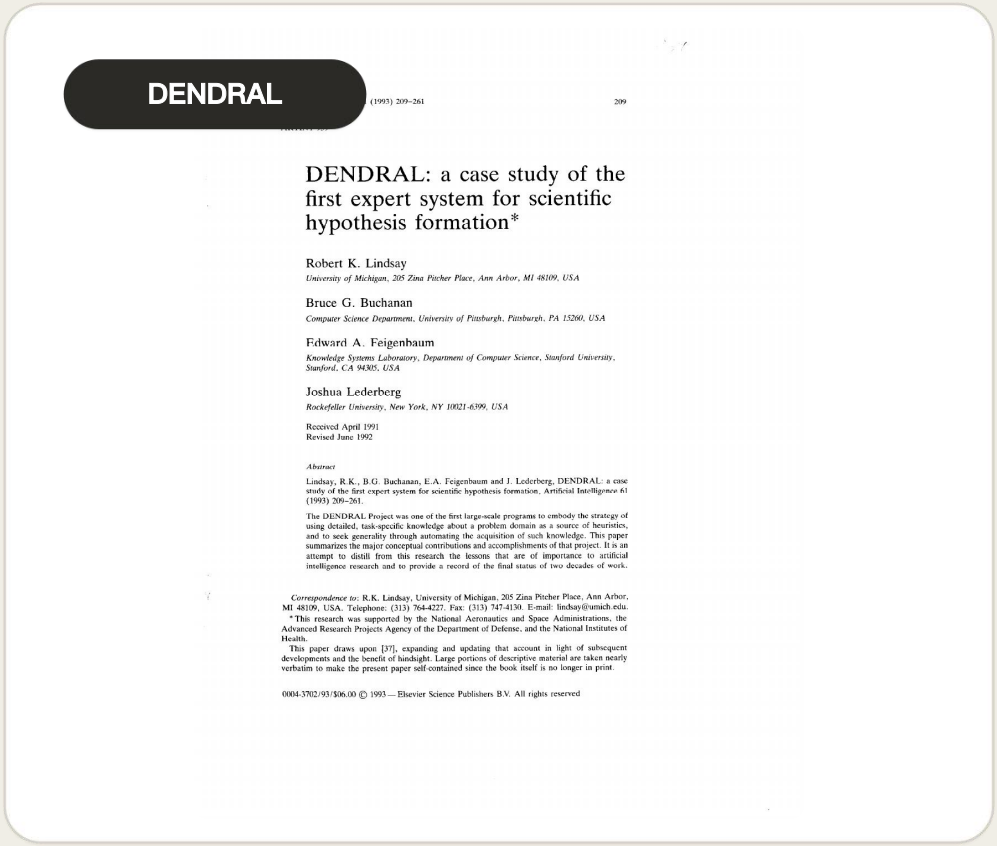
1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点


DENDRAL

(1993) 209–261 209

DENDRAL: a case study of the first expert system for scientific hypothesis formation*

Robert K. Lindsay
University of Michigan, 205 Zina Pitcher Place, Ann Arbor, MI 48109, USA

Bruce G. Buchanan
Computer Science Department, University of Pittsburgh, Pittsburgh, PA 15260, USA

Edward A. Feigenbaum
Knowledge Systems Laboratory, Department of Computer Science, Stanford University, Stanford, CA 94305, USA

Joshua Lederberg
Rockefeller University, New York, NY 10021-6399, USA

Received April 1991
Revised June 1992

Abstract

Lindsay, R.K., B.G. Buchanan, E.A. Feigenbaum and J. Lederberg, DENDRAL: a case study of the first expert system for scientific hypothesis formation, Artificial Intelligence 61 (1993) 209–261.

The DENDRAL Project was one of the first large-scale programs to embody the strategy of using detailed, task-specific knowledge about a problem domain as a source of heuristics, and to seek generality through automating the acquisition of such knowledge. This paper summarizes the major conceptual contributions and accomplishments of that project. It is an attempt to distill from this research the lessons that are of importance to artificial intelligence research and to provide a record of the final status of two decades of work.

Correspondence to: R.K. Lindsay, University of Michigan, 205 Zina Pitcher Place, Ann Arbor, MI 48109, USA. Telephone: (313) 764-4227. Fax: (313) 747-4130. E-mail: lindsay@umich.edu.

* This research was supported by the National Aeronautics and Space Administrations, the Advanced Research Projects Agency of the Department of Defense, and the National Institutes of Health.

This paper draws upon [37], expanding and updating that account in light of subsequent developments and the benefit of hindsight. Large portions of descriptive material are taken nearly verbatim to make the present paper self-contained since the book itself is no longer in print.

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DENDRAL 化学结构分析

MYCIN 感染诊断建议

XCON 计算机配置

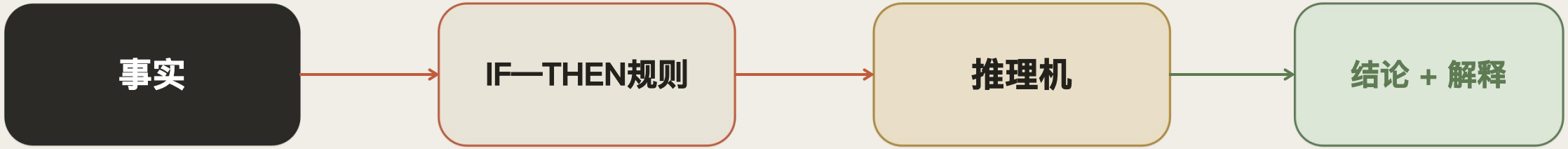
商业逻辑改变

从“造一个通用大脑”转向“复制一个领域专家”。

承上 | 通用智能承诺受挫，神经网络等待条件

下一幕 | 经典规则系统究竟怎样把专家经验变成机器判断10

专家系统的直觉：把“专家怎么判断”写成一条可解释的推理链



优势

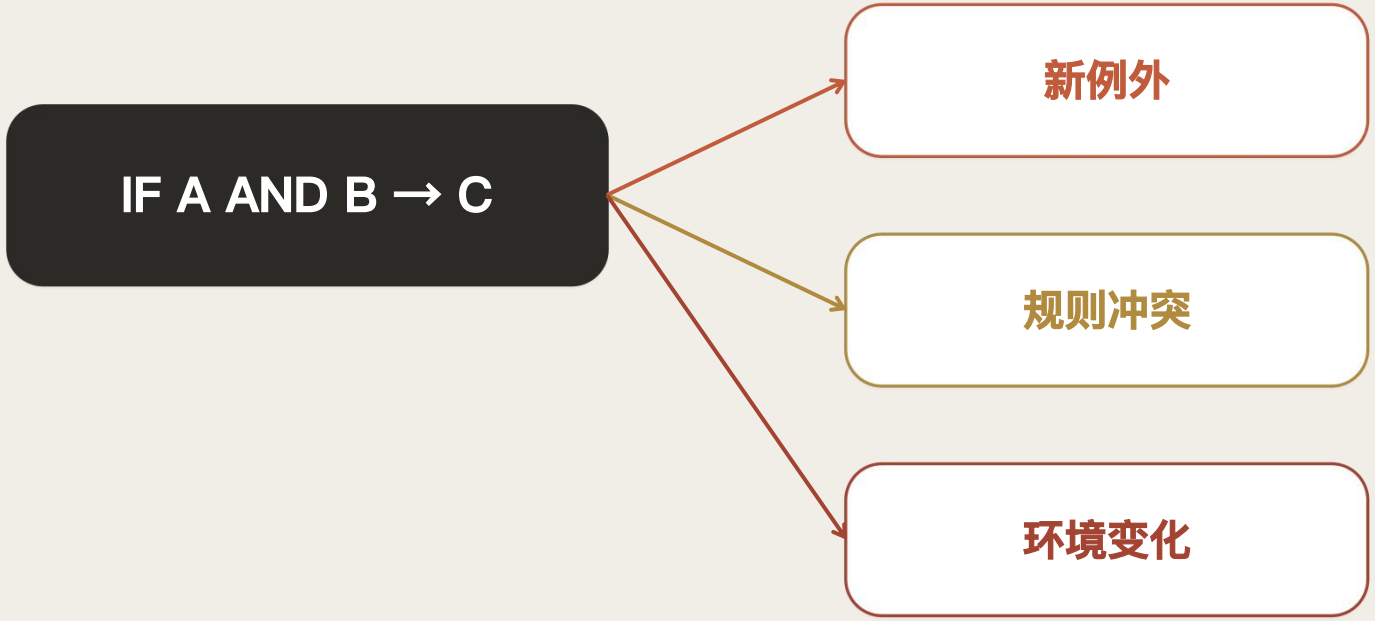
过程可追溯、规则可审核、适合边界明确任务

边界

知识要靠人录入；例外和冲突不断累积；环境变化就要重写

规则没有过时：今天的业务系统、权限边界和高风险动作，仍然需要明确规则。

专家系统的瓶颈不是“不会推理”，而是知识永远写不完、维护不起



商业现实

开发周期长
专家知识难提取
规则维护昂贵
专用硬件市场崩塌

第二次寒冬把问题重新指向学习路线：如果知识写不完，能否让机器从数据里自己归纳?

神经网络重新出发：不再把答案写死，而是让错误反复修改连接

NATURE · 1986



LETTERS TO NATURE 533

...neity of amino acids in fossils. ...d to handle smaller samples, it ...ate individual amino acid enan- ...one enantiomer is entirely derived from the other by racemization during diagenesis, the individual D- and L-enantiomers for a given amino acid should have identical ^{14}C ages.

Older, more poorly preserved fossils may not always prove amenable to the determination of amino acid indigeneity by the stable isotope method, as the prospects for complete replacement of indigenous amino acids with non-indigenous amino acids increases with time. As non-indigenous amino acids undergo racemization, the enantiomers may have identical isotopic compositions and still not be related to the original organisms. Such a circumstance may, however, become easier to recognize as more information becomes available concerning the distribution and stable isotopic composition of the amino acid constituents of modern representatives of fossil organisms. Also, AMS dates on individual amino acid enantiomers may, in some cases, help to clarify indigeneity problems, in particular when stratigraphic controls can be used to estimate a general age range for the fossil in question.

Finally, the development of techniques for determining the stable isotopic composition of amino acid enantiomers may enable us to establish whether non-racemic amino acids in some carbonaceous meteorites[27] are indigenous, or result in part from terrestrial contamination.

M.H.E. thanks the NSF, Division of Earth Sciences (grant EAR-8352055) and the following contributors to his Presidential Young Investigator Award for partial support of this research: Arco, Exxon, Phillips Petroleum, Texaco Inc., The Upjohn Co. We also acknowledge the donors of the Petroleum Research Fund, administered by the American Chemical Society (grant 16144-AC2 to M.H.E., grant 14805-AC2 to S.A.M.) for support. S.A.M. acknowledges NSERC (grant A2644) for partial support.

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Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton† & Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California, San Diego, La Jolla, California 92093, USA
† Department of Computer Science, Carnegie-Mellon University, Pittsburgh, Philadelphia 15213, USA

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal 'hidden' units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure[1].

There have been many attempts to design self-organizing neural networks. The aim is to find a powerful synaptic modification rule that will allow an arbitrarily connected neural network to develop an internal structure that is appropriate for a particular task domain. The task is specified by giving the desired state vector of the output units for each state vector of the input units. If the input units are directly connected to the output units it is relatively easy to find learning rules that iteratively adjust the relative strengths of the connections so as to progressively reduce the difference between the actual and desired output vectors[2]. Learning becomes more interesting but more difficult when we introduce hidden units whose actual or desired states are not specified by the task. (In perceptrons, there are 'feature analysers' between the input and output that are not true hidden units because their input connections are fixed by hand, so their states are completely determined by the input vector: they do not learn representations.) The learning procedure must decide under what circumstances the hidden units should be active in order to help achieve the desired input–output behaviour. This amounts to deciding what these units should represent. We demonstrate that a general purpose and relatively simple procedure is powerful enough to construct appropriate internal representations.

The simplest form of the learning procedure is for layered networks which have a layer of input units at the bottom; any number of intermediate layers; and a layer of output units at the top. Connections within a layer or from higher to lower layers are forbidden, but connections can skip intermediate layers. An input vector is presented to the network by setting the states of the input units. Then the states of the units in each layer are determined by applying equations (1) and (2) to the connections coming from lower layers. All units within a layer have their states set in parallel, but different layers have their states set sequentially, starting at the bottom and working upwards until the states of the output units are determined.

The total input, x_j, to unit j is a linear function of the outputs, y_i, of the units that are connected to j and of the weights, w_{ji}, on these connections

$$x_j = \sum_i y_i w_{ji} \quad (1)$$

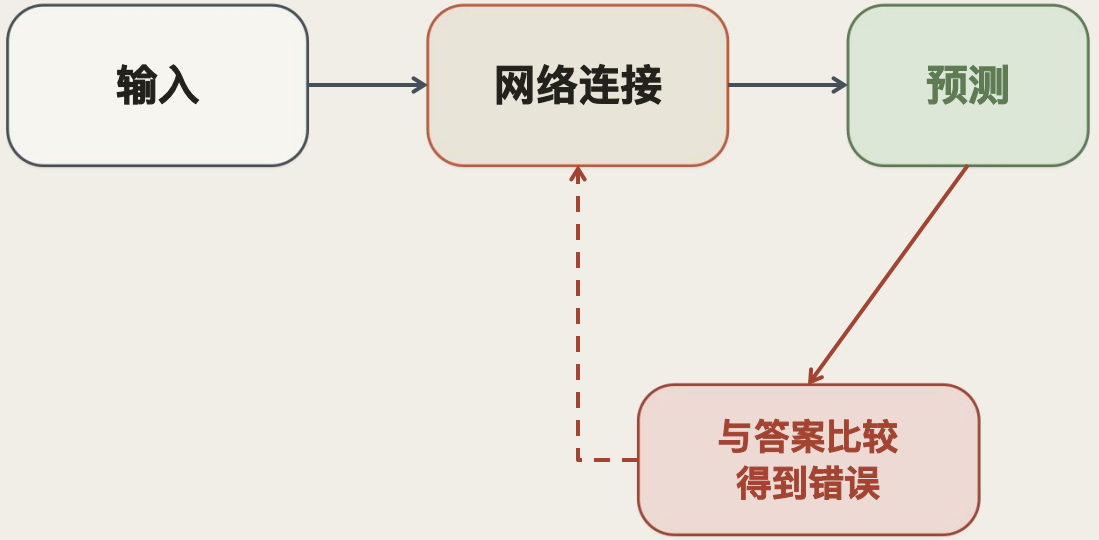
Units can be given biases by introducing an extra input to each unit which always has a value of 1. The weight on this extra input is called the bias and is equivalent to a threshold of the opposite sign. It can be treated just like the other weights.

A unit has a real-valued output, y_j, which is a non-linear function of its total input

$$y_j = \frac{1}{1+e^{-x_j}} \quad (2)$$

† To whom correspondence should be addressed.

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只记住这句话

预测错了 → 把错误向后传 → 调整连接 → 再试一次

灵熙智能内部培训材料

AI开始少谈“复制人类”，多谈“用数据把一个任务做准”

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点

垃圾邮件过滤

从邮件样本学习分类边界

搜索与广告

从点击和相关性数据排序

语音识别

用统计模型处理声音序列

推荐系统

从用户行为预测偏好

这次复兴更务实：明确任务、收集数据、用评测比较改进，而不是直接宣称拥有通用智能。

历史事件 · 1997

Deep Blue击败世界冠军：机器可以在窄域超人，但仍不会跨界

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点



这次胜利组合了旧思想

01 **表示** 把棋盘变成明确状态

02 **搜索** 大规模探索候选走法

03 **知识** 人类棋谱与评估函数

04 **算力** 专用硬件并行计算

历史意义：单项能力可以超越人类；历史边界：会下棋不等于会聊天、迁移或理解开放世界。

条件积累 · 2000s

互联网、数字化与GPU，让学习路线第一次拥有源源不断的燃料

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点

互联网数据
文本、图片、点击、行为

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可规模化的机器学习

GPU并行
图形芯片适合大量矩阵计算

公开评测
ImageNet等基准形成共同目标

历史的方向开始反转：过去是算法等待数据和算力，现在数据与算力开始放大学习算法。

三种经典思想 · 放回历史

搜索、规则、学习没有互相淘汰，而是在现代系统里各自承担角色

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
深度学习拐点

搜索

核心问题

下一步先尝试什么?

今天仍用于

规划、推理、Agent

规则

核心问题

什么必须明确执行?

今天仍用于

权限、流程、安全边界

学习

核心问题

哪些模式无法写完?

今天仍用于

语言、视觉、预测

现代AI的进步更像重新组合历史思想，而不是某一种算法彻底取代其他方法。

第三次浪潮 · 2012

AlexNet点燃深度学习浪潮：不是单点魔法，而是多年条件终于会合

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

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NIPS · 2012



ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
University of Toronto
kriz@cs.utoronto.ca

Ilya Sutskever
University of Toronto
ilya@cs.utoronto.ca

Geoffrey E. Hinton
University of Toronto
hinton@cs.utoronto.ca

Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called "dropout" that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

1 Introduction

Current approaches to object recognition make essential use of machine learning methods. To improve their performance, we can collect larger datasets, learn more powerful models, and use better techniques for preventing overfitting. Until recently, datasets of labeled images were relatively small — on the order of tens of thousands of images (e.g., NORB [16], Caltech-101/256 [8, 9], and CIFAR-10/100 [12]). Simple recognition tasks can be solved quite well with datasets of this size, especially if they are augmented with label-preserving transformations. For example, the current-best error rate on the MNIST digit-recognition task (<0.3%) approaches human performance [4]. But objects in realistic settings exhibit considerable variability, so to learn to recognize them it is necessary to use much larger training sets. And indeed, the shortcomings of small image datasets have been widely recognized (e.g., Pinto et al. [21]), but it has only recently become possible to collect labeled datasets with millions of images. The new larger datasets include LabelMe [23], which consists of hundreds of thousands of fully-segmented images, and ImageNet [6], which consists of over 15 million labeled high-resolution images in over 22,000 categories.

To learn about thousands of objects from millions of images, we need a model with a large learning capacity. However, the immense complexity of the object recognition task means that this problem cannot be specified even by a dataset as large as ImageNet, so our model should also have lots of prior knowledge to compensate for all the data we don't have. Convolutional neural networks (CNNs) constitute one such class of models [16, 11, 13, 18, 15, 22, 26]. Their capacity can be controlled by varying their depth and breadth, and they also make strong and mostly correct assumptions about the nature of images (namely, stationarity of statistics and locality of pixel dependencies). Thus, compared to standard feedforward neural networks with similarly-sized layers, CNNs have much fewer connections and parameters and so they are easier to train, while their theoretically-best performance is likely to be only slightly worse.

1

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01 **多层网络** 自动学习层级图像特征

02 **训练方法** 反向传播与工程技巧

03 **大规模数据** ImageNet标注图像

04 **GPU算力** 大模型训练开始可行

05 **公开评测** 清晰差距形成共同信号

2012之后，AI的主旋律从“人设计特征”转向“网络从数据中学习表征”。

承上 | 搜索、规则、学习在现代系统中重新组合

下一幕 | 从2012走向Agent，历史主线给我们什么判断工具 18

从规则到学习，再到Agent：AI一直在扩大“机器能独立完成的闭环”

1940s—50s
AI成为问题

1956—70s
符号主义兴起

1970s—90s
寒冬与专家系统

1997—2012
机器学习复兴

2012→
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机器学表征

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基础模型
一个模型适配多任务

→

Agent
观察—行动—验证

三条历史判断

① 惊艳演示不等于稳定交付力、反馈和产品共同成熟

② 窄域超人不等于通用智能

③ 突破通常来自方法、数据、算

第二讲的问题：模型如何从“预测下一个token”，走到会写作、推理、调用工具并行动?